DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING IIT (BHU)

Computer Systems Organization (Code: CSO 211)

Time: 24 hours *Mid Term Examination* **Maximum Marks: 30**

All answers should be written in a clear and concise manner. Plagiarism will be checked, and if detected, marks will be deducted. The maximum marks related to the questions are indicated at the end of the questions. Make suitable assumptions for incomplete information.

Q1. (a) Design a combinational circuit with three inputs x, y, z and three outputs A, B, C. When binary input is 0, 1, 2, or 3, the binary output is one greater than input. When binary input is 4, 5, 6 or 7, the binary output is one less than input.

(b) Minimise the following function using K-Maps

F (A, B, C, D) = m (1, 2, 6, 7, 8, 13, 14, 15) + d (0, 3, 5, 12) **[1 + 2 = 3]**

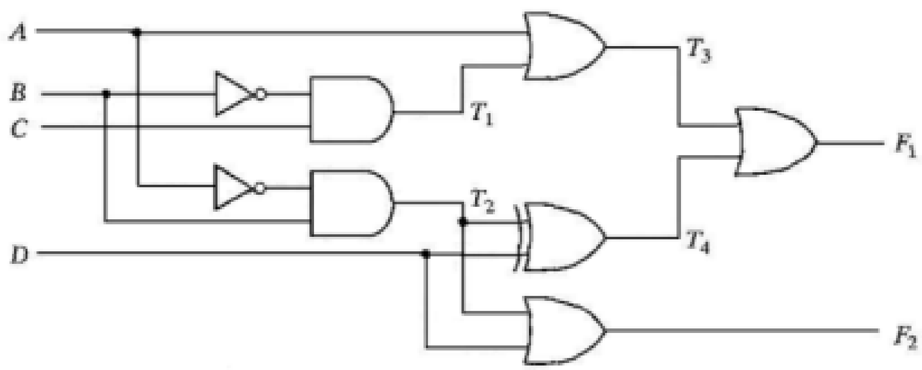
Q2. Represent +60 and - 60 in Sign Magnitude, 1's complement and 2's complement representations using 8 bits. **[1.5 + 1.5 = 3]**

Q3. Consider the combinational circuit shown in Fig.

(a) Derive the Boolean expressions for T1 through T4. Evaluate the outputs of F1 and F2 as a function of the four inputs.

(b) List the truth table with 16 binary combinations of the four inputs variables. Then list the binary values for T1 through T4 and outputs F1 and F2 in the table.

(c) Plot the output Boolean functions obtained in part (b) on maps and show that the simplified Boolean expressions are equivalent to the ones obtained in part (a). **[2 + 2 + 2 =6]**



Q4. (a) Provide the circuit diagram and characteristic table for SR, JK and D flip-flops.

(b) Construct JK and D flip flops from SR flip-flop. **[3 + 2 = 5]**

Q5. What is a decoder? Draw a 2 to 4 decoder circuit using logic gates and also provide the truth table, Boolean expressions. **[1 + 3 = 4]**

Q6. What is a multiplexer? Draw a 8 to 1 multiplexer using logic gates and also provide the truth table, Boolean expression. **[1 + 3 = 4]**

Q7. What are counters? Provide the difference between synchronous and asynchronous counters. Explain the working of 4-bit synchronous and 4-bit asynchronous binary counters. **[1 + 1 + 1.5 + 1.5 = 5]**